In this section, we study with the following steps:

1. Generalized Eigenspace.
2. Cycle of Generalized Eigenvector.
3. Necessary Condition for a given Space to be represented as a Direct sum of Generalized Eigenspaces.
4. Finite-Dimensional Generalized Eigenspace has a basis that represents the linear map as a Jordan block.

0.1 Jordan Canonical Form

Let $T : V \to V$ be a linear operator, and $f(t) \in F[t]$ be a characteristic polynomial of T.
For a linear map $L : V \to V$, and L-invariant subspace $W \leq V$, denote $L|_W$ as a restriction:

$$L|_W : W \to W : x \mapsto L(x)$$

0.1.1 Generalized Eigenspaces

Definition 0.1.1. Suppose that $\lambda \in F$ be an eigenvalue of T. Define a set:

$$K_\lambda \overset{\text{def}}{=} \{x \in V \mid (T - \lambda I)^p(x) = 0 \text{ for some } p \in \mathbb{N}\} = \bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n$$

It is called a *generalized eigenspace* corresponding to λ.
For non-zero $x \in K_\lambda$, the smallest positive integer $p \in \mathbb{Z}$ s.t $(T - \lambda I)^p(x) = 0$ is called *period* of x (or length).
(If $x = 0$, the period of x is defined to be zero.)

Lemma 0.1.2. Let $\lambda \in F$ be an eigenvalue of T. Then, the following properties are satisfied:

1. K_λ is a T-invariant subspace of V containing E_λ.
2. If $\mu \in F$ is an eigenvalue of T such that $\mu \neq \lambda$, then $K_\mu \cap K_\lambda = \{0\}$.

Proof. 1) Let $x, y \in K_\lambda$ and $\alpha \in F$ be given. Denote $p, q \in \mathbb{Z}$ as period of x, y, respectively. Then,

$$(T - \lambda I)^{p+q}(\alpha x + y) = \alpha(T - \lambda I)^{p+q}(x) + (T - \lambda I)^{p+q}(y) = 0.$$

Clearly, $0 \in K_\lambda$. Therefore, subspace criterion gives $K_\lambda \leq V$.
To show T-invariant, let $x \in K_\lambda$. Then, for period $p \in \mathbb{Z}$ of x,

$$(T - \lambda I)^p(T(x)) = ((T - \lambda I)^p \circ T)(x) = (T \circ (T - \lambda I)^p)(x) = T(0) = 0.$$

Thus $T(x) \in K_\lambda$. Including E_λ is clear.

Show 2). Let $w \in K_\mu \cap K_\lambda$ be given. Let $p, q \in \mathbb{Z}$ be period of w for μ and λ, respectively.
Since $\mu \neq \lambda$, $(t - \mu)^p, (t - \lambda)^q \in F[t]$ are disjoint. Now, there exist $g_1(t), g_2(t) \in F[t]$ s.t.

$$g_1(t)(t - \mu)^p + g_2(t)(t - \lambda)^q = 1$$

because $F[t]$ is P.I.D. Now,

$$(g_1(T)(T - \mu I)^p + g_2(T)(T - \lambda I)^q)(w) = 0 + 0 = w$$

Thus $K_\mu \cap K_\lambda = \{0\}$. □

0.1.2 Cycle of generalized eigenvector

Definition 0.1.3. Suppose that $\lambda \in F$ is an eigenvalue of T. For $x \in K_\lambda$ with period $p \in \mathbb{Z}$, Define an ordered set:

$$\mathcal{C}_x \overset{\text{def}}{=} \{(T-\lambda I)^{p-1}(x), (T-\lambda I)^{p-2}(x), \dots, (T-\lambda I)(x), x\}$$

It is called a *Cycle* of generalized eigenvector $x \in V$ corresponding to λ.
In particular, $(T-\lambda I)^{p-1}(x)$ is called *initial vector* and x is called *end vector*.

Note: Clearly, $\langle \mathcal{C}_x \rangle \leq K_\lambda$.

Lemma 0.1.4. Let $\lambda \in F$ be an eigenvalue of T, and $x \in K_\lambda$ be given with period $p \in \mathbb{Z}$. Then,

1. For any $0 \leq i < j \leq p-1$, $(T-\lambda I)^i(x) \neq (T-\lambda I)^j(x)$. That is, the cycle $\mathcal{C}_x$ has p elements.
2. The generated space $\langle \mathcal{C}_x \rangle$ is T-invariant. In fact, it equals to $(T-\lambda I)$-cyclic subspace generated by $x \in V$.
3. The cycle $\mathcal{C}_x$ is linearly independent.
4. If $y \in E_\lambda$ with period $q \in \mathbb{Z}$ satisfies $(T-\lambda I)^{p-1}(x) \neq (T-\lambda I)^{q-1}(y)$, then $\mathcal{C}_x$ and $\mathcal{C}_y$ are disjoint.

Proof. To show 1), Suppose that for some $0 \leq i < j \leq p-1$, $(T-\lambda I)^i(x) = (T-\lambda I)^j(x)$. Take $(T-\lambda I)^{p-j}$ to both side:

$$(T-\lambda I)^{p-j+i}(x) = (T-\lambda I)^p(x) = 0$$

Since $p-(j-i) < p$, this is contradiction to minimality of period. 2) is trivial.
To show 3), suppose that for some $a_i \in F$,

$$\sum_{i=0}^{p-1} a_i (T-\lambda I)^i(x) = 0$$

Then, as take $(T-\lambda I)^{p-1}, (T-\lambda I)^{p-2}, \dots, (T-\lambda I)^1$, we obtain $a_0 = a_1 = \dots = a_{p-1} = 0$.
To show 4), WLOG, assume $p \leq q$ and denote that

$$C_x = \{(T-\lambda I)^{p-1}(x), \dots, x\}, \quad C_y = \{(T-\lambda I)^{q-1}(y), \dots, y\}$$

Suppose that for some $1 \leq i \leq p$ and $1 \leq j \leq q$,

$$(T-\lambda I)^{p-i}(x) = (T-\lambda I)^{q-j}(y)$$

Take $(T-\lambda I)^i$ and $(T-\lambda I)^j$ to both side, then

$$(T-\lambda I)^p(x) = (T-\lambda I)^{q-(j-i)}(y) = 0$$
$$(T-\lambda I)^{p-(i-j)}(x) = (T-\lambda I)^q(y) = 0$$

If $i-j=0$, That is,

$$(T-\lambda I)^{p-i}(x) = (T-\lambda I)^{q-i}(y)$$

Then, as take $(T-\lambda I)^{i-1}$ to both side, we obtain

$$(T-\lambda I)^{p-1}(x) = (T-\lambda I)^{q-1}(y)$$

But this is contradiction to assumption. If $|i-j| > 0$, then contradiction to minimality of period. □

Theorem 0.1.5. Let $\lambda \in F$ be an eigenvalue of T, and $x \in V$ be a generalized eigenvector corresponding to λ. Denote W is a subspace of V generated by the cycle $\mathcal{C}_x$ of x. Then, $[T|_W]_{\mathcal{C}_x}$ is a Jordan-Block.

Proof. Clearly, $[T|_W]_{C_x} = [(T-\lambda I)|_W]_{C_x} + \lambda [I]_{C_x}$ and

$$[(T-\lambda I)|_W]_{C_x} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

gives the result. □

0.1.3 Direct Sum of Generalized Eigenspaces

Lemma 0.1.6. Let $\lambda \in F$ be an eigenvalue of T.
For any $c \in F$ such that $c \neq \lambda$, a restriction $(T - cI)|_{K_\lambda}$ is bijective.

Proof. Since for any $w \in K_\lambda$, $(T - cI)(w) = 0 \implies w \in E_c \subseteq K_c$.
By the Lemma, $w = 0$. That is, kernel of $(T - cI)|_{K_\lambda}$ is zero space, thus injective.
To show surjective, let $x \in K_\lambda$. Since a subspace $\langle \mathcal{C}_x \rangle$ generated by the cycle $\mathcal{C}_x$ of x is $(T - cI)$-invariant, a restriction $(T - cI)|_{\langle \mathcal{C}_x \rangle}$ is injection on finite-dimension space. Thus, it is surjection.
Now, we can find $y \in \langle \mathcal{C}_x \rangle$ s.t $(T - cI)(y) = x$. □

Lemma 0.1.7. If $f(t)$ splits over F, then for any eigenvalue λ with multiplicity $m \in \mathbb{N}$,

$$\dim K_\lambda \leq m \quad \text{and} \quad K_\lambda = \ker((T - \lambda I)^m)$$

Proof. Let $h(t)$ be a characteristic polynomial of a restriction $T|_{K_\lambda}$.
Since for any distinct eigenvalue $\mu \in F$, $K_\mu \cap K_\lambda = \{0\}$, λ is a unique eigenvalue of $T|_{K_\lambda}$. Because $h(t)$ divides $f(t)$,

$$h(t) = (t - \lambda)^d$$

where $d \leq m$. Now, show the second assertion. Clearly, $\ker(T - \lambda I)^m \subseteq K_\lambda$.
Meanwhile, by assumption, $f(t) = (t - \lambda)^m \cdot g(t)$ where $g(\lambda) \neq 0$.
Therefore, $g(T)|_{K_\lambda}$ is a bijection, by the Lemma. Now, for any $x \in K_\lambda$, there is $y \in K_\lambda$ s.t. $g(T)(y) = x$.
The Cayley-Hamilton Theorem gives

$$(T - \lambda I)^m(x) = (T - \lambda I)^m(g(T)(y)) = f(T)(y) = T_0(y) = 0.$$

Thus $x \in \ker(T - \lambda I)^m$. □

Theorem 0.1.8. Let V be a finite dimensional vector space, $T : V \to V$ be a linear operator.
If a characteristic polynomial $f(t)$ of T splits over F, then for distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ of T,

$$V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}.$$

Proof. Denote $n = \dim V$ and

$$f(t) = (-1)^n (t - \lambda_1)^{n_1} \ldots (t - \lambda_k)^{n_k}.$$

And for each $1 \leq j \leq k$, define

$$f_j(t) = \prod_{\substack{i \neq j \\ 1 \leq i \leq k}} (t - \lambda_i)^{n_i}$$

Since the greatest common divisor for $f_j \ (1 \leq j \leq k)$ is only 1, there exist $q_j(t) \ (1 \leq j \leq k)$ such that

$$q_1(t)f_1(t) + \cdots + q_k(t)f_k(t) = 1.$$

Now, for any $x \in V$,

$$q_1(T)f_1(T)(x) + \cdots + q_k(T)f_k(T)(x) = x.$$

Meanwhile for each $q_j(T)f_j(T)(x)$, the Cayley-Hamilton Theorem gives

$$(T - \lambda_j I)^{n_j}(q_j(T)f_j(T)(x)) = (T - \lambda_j I)^{n_j}(f_j(T)q_j(T)(x)) = f(T)(q_j(T)(x)) = 0$$

That is, $q_j(T)f_j(T)(x) \in K_{\lambda_j}$. This proved $V = K_{\lambda_1} + \cdots + K_{\lambda_k}$.
To show the Uniqueness, note that: since K_{λ_j} is T-invariant, so it is $f_j(T)$-invariant.
Also, a restriction $f_j(T)|_{K_{\lambda_j}}$ is bijection, being composition of bijections.
Moreover, for $i \neq j$, $f_j(T)[K_{\lambda_i}] = f_j(T)[\ker(T - \lambda_i I)^{n_i}] = \{0\}$.
Now, if $\sum_{i=1}^k v_i = \sum_{i=1}^k w_i$ where $v_i, w_i \in K_{\lambda_i} \ (1 \leq i \leq k)$, then for each $1 \leq j \leq k$,

$$f_j(T)(v_j) = \sum_{i=1}^k f_j(T)(v_i) = \sum_{i=1}^k f_j(T)(w_i) = f_j(T)(w_j).$$

Since $f_j(T)|_{K_{\lambda_j}}$ is injection, $v_j = w_j$. □

0.1.4 The Existence of Jordan-Canonical Form

Now, we know that: if the characteristic polynomial of T splits over F, then the space V can be represented by direct sum of generalized eigenspaces. And, the cycle of eigenvector generates some subspace of generalized eigenspace, moreover, it is Jordan-basis for the restriction. If we can show that some cycles form a partition for basis of the eigenspace, finally, we can conclude that there exists a Jordan Canonical Form.

Lemma 0.1.9. Let $T : V \to V$ be a linear operator, and $\lambda \in F$ be an eigenvalue of T.
Consider that the generalized eigenvectors $x_1, \dots, x_q \subset K_\lambda$ with period $p_i \in \mathbb{Z}$, respectively.
If elements in $\{(T - \lambda I)^{p_i - 1}(x_i) \mid 1 \le i \le q\}$ are mutually different and the set is linearly independent, then

$$\mathcal{C} \overset{\text{def}}{=} \bigcup_{i=1}^{q} \mathcal{C}_{x_i}$$

is linearly independent. (Disjointness is already shown by the Lemma.)

Proof. Using mathematical induction for number of elements of $\mathcal{C}$.
If $\mathcal{C}$ has only one element, then trivial. Suppose that the statement holds for any $\mathcal{C}$ having fewer than n elements.
Now, suppose that $\mathcal{C}$ has exactly n elements. Denote that

$$W \overset{\text{def}}{=} \langle \mathcal{C} \rangle, \;\; U \overset{\text{def}}{=} (T - \lambda I)|_W$$

Since each $\mathcal{C}_{x_i}$ generates T-invariant subspace and $\langle A \cup B \rangle = \langle A \rangle + \langle B \rangle$ and sum of T-invariant is T-invariant,
W is T-invariant, also $(T - \lambda I)$-invariant. Thus, the restriction U is well-defined.
For each $1 \le i \le q$, $\mathcal{C}_{x_i} \setminus \{x_i\}$ is either emptyset or a cycle having end vector as $(T - \lambda I)(x_i)$, having same initial vector.

$$\operatorname{im} U = U[\langle \mathcal{C} \rangle] = U \left[\left\langle \bigcup_{i=1}^{q} \mathcal{C}_{x_i} \right\rangle \right] = U \left[\sum_{i=1}^{q} \langle \mathcal{C}_{x_i} \rangle \right] = \sum_{i=1}^{q} U \left[\langle \mathcal{C}_{x_i} \rangle \right] = \sum_{i=1}^{q} \langle U \left[\mathcal{C}_{x_i} \right] \rangle = \sum_{i=1}^{q} \langle \mathcal{C}_{x_i} \setminus \{x_i\} \rangle = \left\langle \underbrace{\bigcup_{i=1}^{q} \mathcal{C}_{x_i} \setminus \{x_i\}}_{\mathcal{C}'} \right\rangle$$

Since the set $\mathcal{C}'$ is consisted by $n - q$ elements, and the initial vectors are linearly independent,
therefore $\mathcal{C}'$ is linearly independent, by hypothesis of the induction. Thus it is a basis for $\operatorname{im} U$.
Moreover, $\ker U$ has more than q elements, because each initial vector of $\mathcal{C}_{x_i}$ is contained in $\ker U$. Consequently,

$$n \ge \dim W = \dim \operatorname{im} U + \dim \ker U \ge (n - q) + q = n$$

This implies $\dim W = n$, thus $\mathcal{C}$ is basis of W. Thus, linearly independent. □

> **Theorem 0.1.10.** Let $T : V \to V$ be a linear operator, and $\lambda \in F$ be an eigenvalue of T.
> If the generalized eigenspace K_λ has finite dimension,
> then it has a basis consisting of a disjoint union of cycles of generalized eigenvectors.

Proof. Using mathematical induction for dimension of K_λ. If $\dim K_\lambda = 1$, trivial.
Suppose that for $n \in \mathbb{N}$, the statement holds for any generalized eigenspace K such that $\dim K < n$.
Now, assume that $\dim K_\lambda = n$. Denote $U \overset{\text{def}}{=} (T - \lambda I)|_{K_\lambda}$. Note that: $\operatorname{im} U < K_\lambda$ because the Dimension Theorem gives

$$\dim K_\lambda = \dim \operatorname{im} U + \dim \ker U$$

But $\ker U = E_\lambda \leq K_\lambda$ is not a zero space, $\dim \operatorname{im} U < \dim K_\lambda$.
Moreover, for any $x \in \operatorname{im} U$, $x = (T - \lambda I)(y)$ for some $y \in K_\lambda$, by definition. Now

$$T(x) = T((T - \lambda I)(y)) = (T - \lambda I)(T(y)) \in \operatorname{im} U$$

That is, $\operatorname{im} U$ is T-invariant, which means $\operatorname{im} U$ is generalized eigenspace of $T|_{\operatorname{im} U}$ corresponding to λ.
By the assumption of induction, there exist disjoint cycles $\mathcal{C}_i,\ 1 \leq i \leq q$ such that

$$\mathcal{C} = \bigcup_{i=1}^{q} \mathcal{C}_i$$

is a basis of $\operatorname{im} U$. For each $1 \leq i \leq q$, the end vector of $\mathcal{C}_i$ is contained in $\operatorname{im} U$, thus we can extend the cycle $\mathcal{C}_i$ to

$$\overline{\mathcal{C}}_i = \mathcal{C}_i \cup \{v_i\}$$

And, denote w_i as an initial vector of a cycle $\mathcal{C}_i$(It is also initial vector of $\overline{\mathcal{C}}_i$.)
Since the initial vectors are contained in E_λ and linearly independent, we can obtain extended basis for E_λ,

$$\{w_1, \ldots, w_q, u_1, \ldots, u_s\}$$

Finally, initial vectors in the following cycles are linearly independent:

$$\overline{\mathcal{C}}_1, \ldots, \overline{\mathcal{C}}_q, \{u_1\}, \ldots, \{u_s\}$$

by the Lemma, the union of these sets is linearly independent. The Dimension Theorem gives the result. □